

SAMPREETH AVVARI

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SUMMARY

Software Engineer who designs and ships reliable backend and full-stack services end to end, from design through deployment and on-call. Built 5 production systems solo and a pipeline handling 500K+ daily events, cutting deployment errors 40%.

EDUCATION

MS, Computer Engineering, New York University, NY (CGPA 3.84/4)

Aug 2023 - May 2025

BTech, Computer Science and Engineering, JNTU, India (CGPA 3.92/4)

Jul 2017 - Jun 2021

SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, C++, SQL

Backend & Systems: FastAPI, Flask, Node.js, REST, GraphQL, microservices, event-driven architecture, distributed systems, low-latency services, caching, Pydantic/Zod

Data & Cloud / DevOps: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, BigQuery, Kafka, Spark; GCP (Cloud Run, Pub/Sub), AWS (EC2, S3, RDS, MSK), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Jenkins), monitoring and observability (Prometheus, Grafana, OpenTelemetry)

Frontend & AI: React, Next.js, Tailwind, Drizzle ORM, Vitest; LLM integration, RAG, Vertex AI Gemini, prompt engineering

Foundations: data structures and algorithms, system design, testing (pytest, Selenium), code review, cross-functional collaboration, on-call ownership

WORK EXPERIENCE

Software Engineer | Hybride Implants, New York

Sep 2025 - Present

- Partnered with non-technical stakeholders to build and operate 5 production services solo as sole engineer and on-call owner, shipping each from design through deployment at under \$50 a month.
- Built shared REST and Pub/Sub messaging contracts with Pydantic validation across 5 loosely-coupled, independently deployable microservices.

CBCT Scan Validator, dental-CT validation service

- Designed a high-throughput, event-driven CT-scan validation service (Pub/Sub, OpenVINO) processing 600 scans a month at 5.5s each, fault-tolerant and scale-to-zero.
- Owned the ingestion-to-serving pipeline (SFTP, parity-checked ETL) on CPU-only Cloud Run, gating deploys on a 20-scan holdout before promotion to production.

Treatment Estimator, Next.js and Postgres quoting platform

- Delivered a Next.js and Postgres quoting platform with database-enforced invariants (write-once columns, triggers) and 100% branch test coverage, serving 50 consults a week.
- Replaced a decade-old vendor tool with a guided coordinator workflow and a 5-model pricing engine behind a maintainable, well-tested service.

Doc Coach, LLM consultation-QA microservice

- Shipped an LLM consultation-QA microservice (FastAPI, Cloud Run) with JSON Schema contracts and one-retry error handling, scoring every consult against a 7-criterion rubric.

Software Engineer | Shure Incorporated, India

Aug 2021 - Aug 2023

- Engineered a distributed, event-driven ingestion service on AWS (Kafka, MSK, DynamoDB, S3) handling 500K+ daily events with async workers, cutting latency 20% at high throughput.
- Directed delivery across 3 release tracks with Jenkins canary deploys and rollback, cutting deployment errors 40% and debugging production issues with 150+ pytest and Selenium tests at 90%+ coverage.
- Led code reviews across 3 services and authored the Factory and Strategy test framework that cut manual QA 70%.

RESEARCH EXPERIENCE

Machine Learning Researcher | New York University (writeup)

May 2024 - Sep 2025

- Engineered a Python and Airflow ETL turning 118 Reddit CMV shards into 38K examples with schema enforcement, failure alerting, and checkpointed recovery, to fine-tune Llama 3.1 8B.

PROJECTS

JobPilot, open-source (MIT) job-hunt autopilot

- Architected JobPilot, an open-source autopilot (12 users) with a plug-in source architecture: 11 ATS connectors behind one schema, scanning hundreds of postings a day for under \$10 a month.
- Designed a Sheets-as-database, a self-refining rewrite loop, and idempotent per-source fetchers that fail independently, shipped via GitHub Actions CI/CD with 50+ tests.

Loan Radar, self-retraining loan-default scoring platform

- Packaged Loan Radar, a self-retraining loan-default platform on Terraform Kubernetes (Ray), as 3 FastAPI microservices with Docker and MLflow-tracked CI/CD for rollback-safe deploys.